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10.3.4 Interxylary or Included Phloem

The secondary phloem found in the form of strands (islands) embedded in the secondary xylem is called **interxylary** or **included phloem**.

Example: *Strychnos, Combretum* and *Salvadora*.

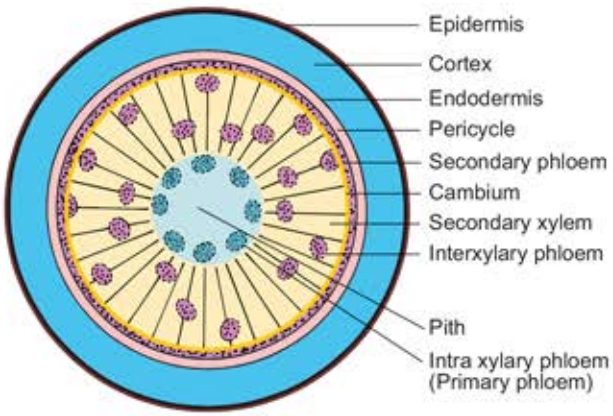


Figure 10.21: Transverse section (T.S) of *Strychnos* stem (diagrammatic) showing anomalous secondary growth

10.3.5 Presence of medullary bundles along with normal cambial activity

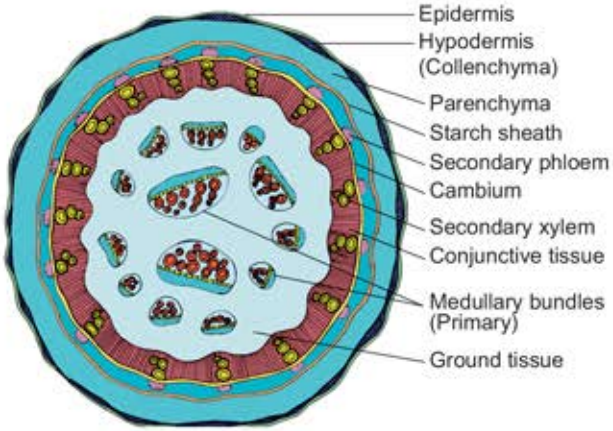


Figure 10.22: Transverse section (T.S) of Boerhaavia stem (diagrammatic) showing anomalous secondary growth

In this type, the normal vascular bundles are arranged in a ring from which a normal cambial ring is formed. In addition, vascular bundles are scattered or found in a ring in the pith or medullary region. These are called **medullary vascular bundles**. Example: *Boerhaavia*.

10.3.6 Presence of cortical bundles along with normal cambial activity

In this type, a ring of vascular bundles are found, from which a normal cambial ring is formed. In addition, in the cortical region additional vascular bundles are found. Example: *Nyctanthes*.

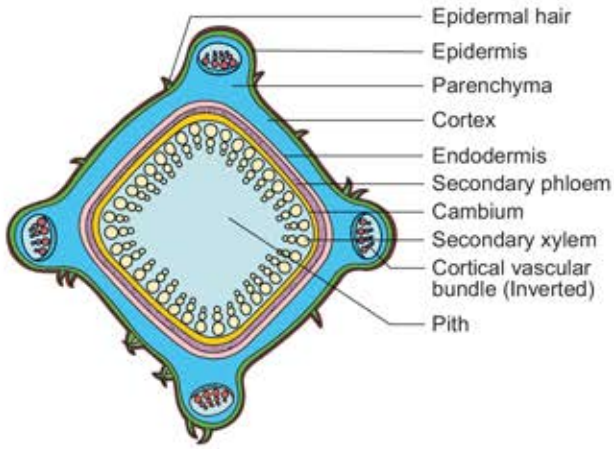


Figure 10.23: Transverse section (T.S) of *Nyctanthes* stem (diagrammatic) showing anomalous secondary growth

10.3.7 Intraxylary or Internal Phloem

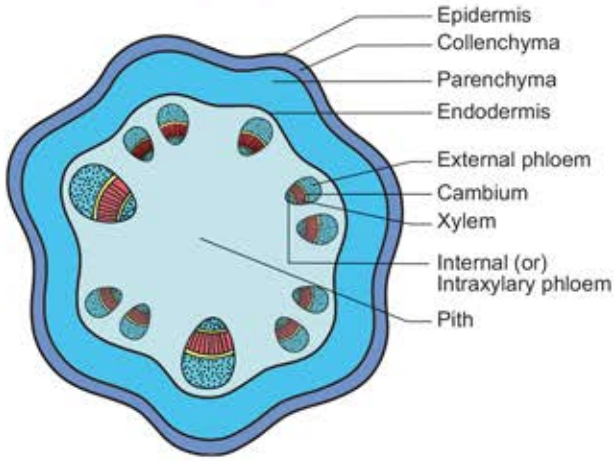


Figure 10.24: Transverse section (T.S) of *Solanum tuberosum* stem (diagrammatic) showing anomalous secondary growth

The cambium usually forms primary phloem on the outer side. In some plants,it